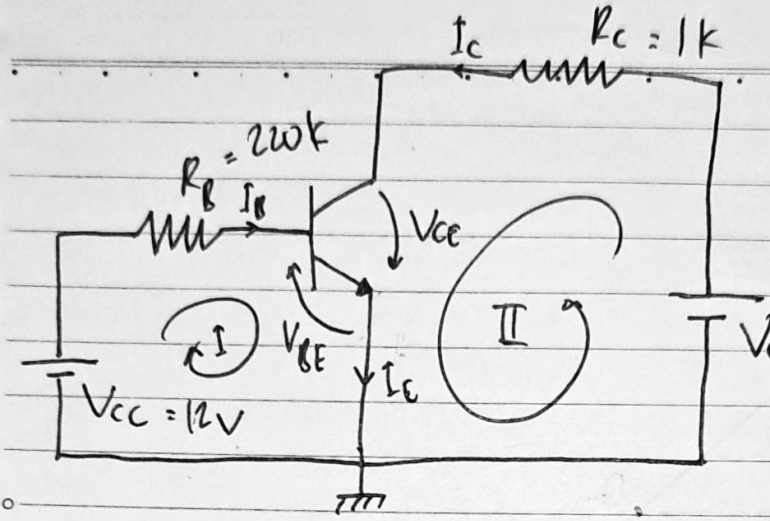


$\beta = 100$

Persamaan :

Loop I → $-V_{CC} + I_B \cdot R_B + V_{BE} = 0$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0,7}{220 \times 10^3} = 0,051 mA = 51 \mu A$$

Loop II → $-V_{CC} + I_C \cdot R_C + V_{CE} = 0$

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

$V_{CE} = V_{CC} - I_C \cdot R_C$; $\beta = \frac{I_C}{I_B}$ → $I_C = \beta \cdot I_B = 100 \times 51 \times 10^{-6}$

$= 5,1 mA$

$V_{CE} = V_{CC} - \beta \cdot I_B \cdot R_C$

$= 12 - 100 \cdot 51 \times 10^{-6} \cdot 1000$

$= 6.9 V$

I_C saturasi = $V_{CE} = 0$

$-V_{CC} + I_C \cdot R_C = 0$

$$I_C = \frac{V_{CC}}{R_C}$$

$$= \frac{12}{1000} = 12mA$$

$$\alpha_{dc} = \frac{I_E}{I_C} \; ; \; I_E = I_B + I_C$$

$$= \frac{I_B + I_C}{I_C}$$

$$= \frac{\beta + 1}{\beta} = \frac{100 + 1}{100} = 1,01$$

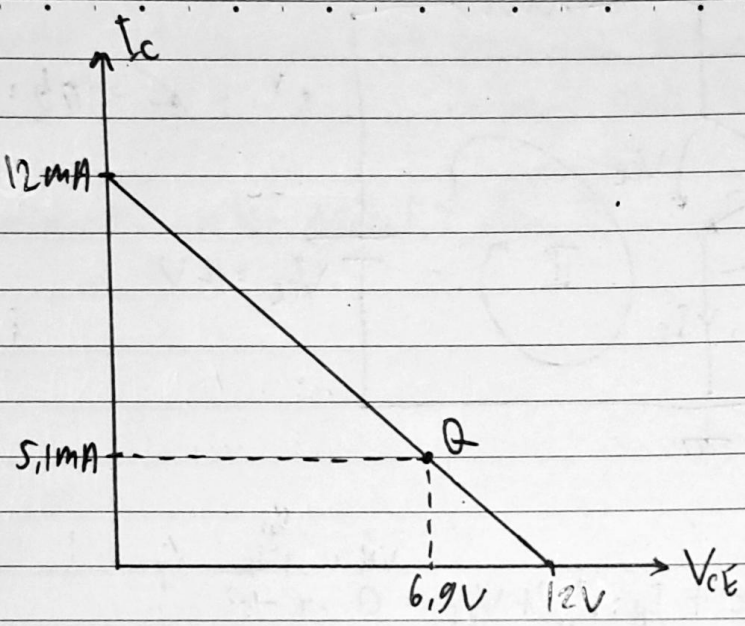
V_{CE} cut off → $I_C = 0$

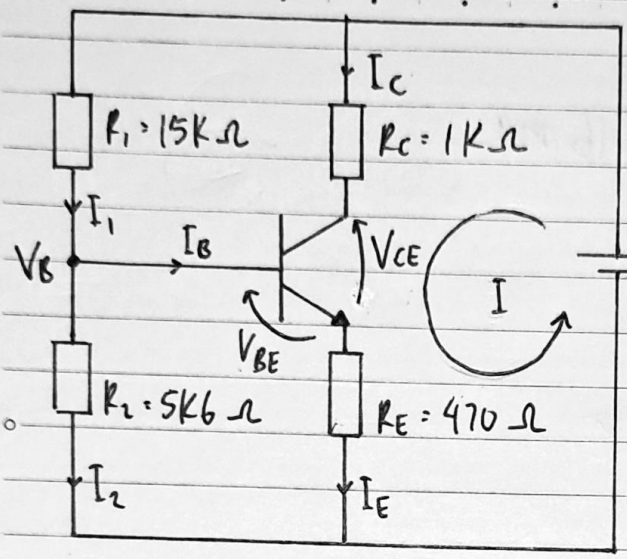
$-V_{CC} + V_{CE} = 0$

$V_{CE} = V_{CC}$

$= 12V$

KENKO 30 Lines, 6 mm


I_C
12mA
5,1mA
Q
6,9V
12V
V_{CE}



$I_1 = I_2 + I_B$ $\quad \alpha = \frac{I_E}{I_C}$

$I_E = I_B + I_C$

$V_{CC} = 12\,V$ $\quad \beta = \frac{I_C}{I_B}$

$\beta = 100$

$V_{BE} = 0{,}7\,V$

$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = 3{,}26\,V.$

$V_{BE} = V_B - V_E$

$V_E = V_B - V_{BE}$

$= 3{,}26 - 0{,}7$

$= 2{,}56\,V.$

$V_E = I_C \cdot R_E$

$I_C = \frac{V_E}{R_E} = \frac{2{,}56}{470} = 5{,}45\,MA$

LOOP I :

$V_{CC} = I_C \cdot R_C + V_{CE} + V_E$

$= I_C \cdot R_C + V_{CE} + I_C \cdot R_E$

$= I_C (R_C + R_E) + V_{CE}$

$V_{CE} = V_{CC} - I_C (R_C + R_E)$

$= 12 - 5{,}45 \times 10^{-3} (1000 + 470)$

$= 3{,}98\,V$

$\beta = \frac{I_C}{I_B}$

$I_B = \frac{I_C}{\beta} = \frac{5{,}45 \times 10^{-3}}{100} = 54{,}5\,\mu A$

$I_E = I_B + I_C$

$= 54{,}5 \times 10^{-6} + 5{,}45 \times 10^{-3}$

$= 5{,}5\,MA$

$\alpha dc = \frac{I_E}{I_C} = \frac{I_B + I_C}{I_C} = \frac{\beta + 1}{\beta}$

$= \frac{5{,}5 \times 10^{-3}}{5{,}45 \times 10^{-3}}$ $\quad = \frac{100 + 1}{100}$

$= 1{,}01$ $\quad = 1{,}01$

No
Date

I_C saturasi $\longrightarrow V_{CE} = 0$

$V_{CC} = I_C (R_C + R_E) + V_{CE}$

$I_C = \frac{V_{CC}}{R_C + R_E} = \frac{12}{100 + 470} = 8{,}16 \text{ mA}$

V_{CE} cut off $\longrightarrow I_C = 0$

$V_{CC} = I_C (R_C + R_E) + V_{CE}$

$V_{CC} = V_{CE}$

$12V = V_{CE}$

